

$Z\phi$ -classes for PerfectGroup(1920, 3)

Claude Archer

$$Z(P) \cong C_2, \quad \text{Out}(P) \cong \text{Sym}(4)$$

n	Runtime (s)	$\#Z\phi$	Cumulative
2	0.011	6	6
3	0.019	2	8
4	12.277	34	42
5	0.013	1	43
6	0.150	16	59
7	0.012	1	60
8	1.821	196	256
9	0.065	4	260
10	0.163	12	272
11	0.012	1	273
12	1.352	116	389
13	0.019	1	390
14	0.299	12	402
15	0.036	2	404
16	13.399	1,606	2,010
17	0.013	1	2,011
18	1.086	42	2,053
19	0.011	1	2,054
20	1.174	104	2,158
21	0.122	4	2,162
22	0.155	12	2,174

n	Runtime (s)	$\#Z\phi$	Cumulative
23	0.013	1	2,175
24	9.160	848	3,023
25	0.051	2	3,025
26	0.192	12	3,037
27	0.602	12	3,049
28	1.129	96	3,145
29	0.016	1	3,146
30	1.030	32	3,178
31	0.013	1	3,179
32	115.726	21,624	24,803
33	0.037	2	24,805
34	0.264	12	24,817
35	0.031	1	24,818
36	12.437	382	25,200
37	0.014	1	25,201
38	0.153	12	25,213
39	0.111	4	25,217
40	8.577	830	26,047
41	0.016	1	26,048
42	1.184	48	26,096
43	0.013	1	26,097

n	Runtime (s)	$\#Z\phi$	Cumulative
44	1.189	96	26,193
45	0.161	4	26,197
46	0.210	12	26,209
47	0.014	1	26,210
48	93.585	9,568	35,778
49	0.046	2	35,780
50	0.897	30	35,810
51	0.034	2	35,812
52	1.390	104	35,916
53	0.012	1	35,917
54	5.958	138	36,055
55	0.104	2	36,057
56	12.317	743	36,800
57	0.132	4	36,804
58	0.301	12	36,816
59	0.014	1	36,817
60	7.322	374	37,191
61	0.013	1	37,192
62	0.281	12	37,204
63	0.511	9	37,213

Summary. For $2 \leq n \leq 63$, the cumulative number is **37,213** $Z\phi$ -classes. The measured total runtime is **307.469 s** (about 5 min 7.5 s). Each row counts the $Z\phi$ -classes attached to auxiliary groups S of order $2n$; the cumulative column starts at $n = 2$. Runtimes are empirical GAP timings for the validated v7 implementation.

Runtime concentration for PerfectGroup(1920, 3)

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$$Z(P) \cong C_2, \quad \text{Out}(P) \cong \text{Sym}(4)$$

n	Runtime (s)	$\#Z\phi$ -classes	Share of total runtime	Cumulative share
32	115.726	21,624	37.64%	37.64%
48	93.585	9,568	30.44%	68.08%
16	13.399	1,606	4.36%	72.43%
36	12.437	382	4.04%	76.48%
56	12.317	743	4.01%	80.48%
4	12.277	34	3.99%	84.48%
24	9.160	848	2.98%	87.46%
40	8.577	830	2.79%	90.25%
60	7.322	374	2.38%	92.63%
54	5.958	138	1.94%	94.56%
8	1.821	196	0.59%	95.16%
52	1.390	104	0.45%	95.61%

Reading the timing profile. The computation is highly concentrated. The two cases $n = 32$ and $n = 48$ alone account for **68.08%** of the total runtime, while the ten slowest values account for **94.56%**. Thus most values of n are computationally negligible compared with a small set of even orders.

Structural interpretation. The largest costs occur when the auxiliary groups S (of order $2n$) have rich normal-subgroup lattices and/or large elementary-abelian sections, producing many admissible pairs before the action of $\text{Aut}(S)$ fuses them into comparatively few orbits. This phenomenon was visible, for example, at $n = 32$ in groups such as $C_4 \times C_2^4$, and at $n = 48$ in $C_3 \times C_2^5$ and $C_2^4 \times S_3$. The validated v7 algorithm already exploits an elementary-abelian fast path and caches quotient packets; the remaining dominant costs are mainly `NormalSubgroups(S)`, `AutomorphismGroup(S)`, and the construction/action on admissible normal subgroups.

Scope. This table deliberately stops at $n = 63$. The case $n = 64$ (auxiliary groups of order 128) is kept separate because it contains 2,328 groups and merits its own audited run. The values displayed here are the complete results for $2 \leq n \leq 63$ from the validated v7 computation.